

Notes from Week 1;

TCP/IP: The TCP/IP created at 1976 in DARPA

It describes the functions performed in particular layers by the popular set of network protocols

OSI/ISO model

It describes all stages of communication in various computer networks

Encapsulation:

Sending data from higher layer to lower layer

Decapsulation:

Decomprising data lower to higher one

Network protocols can be implemented in hardware, software both simultaneously

Most Important part

OSI

TCP/IP

Applications				HTTP, SQL, DHCP DNS, FTP, POP, SMTP	Application
Presentations					
Sessions					
Transport	TCP segment UDP datagram	Port	Router	TCP, UDP, SCTP	Transport
Network	Packet	IP	Switch	IPv4, IPv6, ICMP	Internet
Data Link	Frame	MAC	Hub	Ethernet, ARP, PPP CDP	Network Access
Physical				Bluetooth, DSL, 802.11	

DNS → domain name servers → TCP/UDP 53

SMTP → Simple mail Transfer Protocol → TCP 25

POP → Post Office → TCP port 110

IMAP → Internet Message Access Protocol TCP 143

DHCP → Dynamic Host Configuration " UDP → 67

FTP → File Transfer Protol 20 & 21

SSH → Secure Shell - TCP port 22

Telnet → TCP port 23

HTTP → 80

HTTPS → 443

PostgreSQL → 5432

MySQL → 3306

Physical layer:

Is responsible for physical transport with network media in the form of a sequence of bits

Physical layer protocols define:

Physical properties of data carrier

Coding binary values in the medium (modulation)

It does not include ways of verifying data integrity

Data Link Layer:

Packs the data into frames and sends it to physical layer

CRC control of sent messages

Sender and recipient ID on the transmission medium.

MAC address

Address of computers in local network

24 bits for manufacturer

24 bits for identifier

Network layer:

Finding best route for a packet in the network

Transport layer:

Ensuring that the data sent will be received by the appropriate application on the recipient's side

Responsible for dividing into segments and merging them

TCP Segment / UDP datagram

TCP: A connection-oriented protocol that guarantees reliable, in order delivery of data used for applications where accuracy is critical such as web browsing

UDP:

A connectionless protocol that prioritizes speed over reliability.

Session layer:

Supervises the connection between the source and target

Presentation layer

Compression and decompression of data transferred from the sender to the recipient

Application layer

Enables direct interaction of user applications on the network